CIS*4900: Using Natural Language Processing to Analyze Medical Debt in Civil Court Data

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Abstract

The Civil Justice Data Commons, developed and hosted by the Massive Data Institute and Georgetown University Law Center, has a large collection of civil court data. In this study, civil court data from that repository was used with the goal of classifying plaintiffs as medical or non-medical to understand and analyze medical debt in the US. In this study, the classification of plaintiffs as medical or non-medical was performed in two ways: the first classification method was basic keyword filtering, and the second was to use word embeddings— real-number vectors that represent words from a vocabulary— to flag words in plaintiff data related to the medical field. The effect of various word embedding models is explored, and the frequency and accuracy of medical flag words is assessed.

I. Introduction

The Civil Justice Data Commons (CJDC) is a large collection of civil court data developed and hosted by the Massive Data Institute and Georgetown University Law Center so that people may better understand the civil legal system in the United States [4]. In this study, civil court data from the CJDC was used to understand medical debt in the US by analyzing the plaintiffs bringing suits to court and labelling these plaintiffs as medical or non-medical related. Determining whether the string representing the plaintiff contains medical-related substrings is therefore of great value.

In this paper, both common and machine knowledge was used to create sets of terms deemed “medical-related” against which the plaintiffs were checked. An analysis of the specific words that flagged each plaintiff as medical can then be used to evaluate the efficacy of the classification systems.

This paper is structured as follows: section II discusses the civil court data which the study aims to classify as medical or not, section III describes how the medical classification was conducted, and the results of such classification is discussed in section IV. Finally, section V concludes the paper and provides direction for more accurate labelling of this data.

II. Data

The major source of this data was through the Civil Justice Data Commons, a repository for civil legal data gathered from courts, legal service providers, and other civil law institutions [4]. Four datasets were provided for this study from civil courts in Philadelphia, PA, Maricopa County, AZ, Oklahoma, and Maryland; however, the former two datasets contained solely eviction data. Eviction, the removal of a tenant from rental property by the landlord, does not contribute to medical debt, therefore only the Oklahoma and Maryland datasets will be used and discussed in this paper.

The Oklahoma dataset used, Oklahoma full_INTERNAL, has over 3 million records, though only one table in this dataset, "OJO Civ Cases", was used. This table has only a fraction of its parent's records, as it contains 762511 rows and 14 columns. A portion of the data in the "OJO Civ Cases" table can be found in the below Table I, though for the purposes of this study only the data in the plaintiff column was examined; as this is a preliminary dive into the data, the first focus of the study was conducted on those who brought the case to court. In the Oklahoma dataset, the data in the plaintiff columns consists of uppercase strings corresponding to the name of the case's plaintiff.

Like the Oklahoma dataset, the Maryland dataset used, Maryland CLUE_uncleaned_INTERNAL, has over 3 million entries, though it has only one table, "Cases". This table, modelled in table II, has 3,253,447 rows and 28 columns, including one corresponding to the case's plaintiff. However, where the plaintiff column of the Oklahoma dataset was already parsed into an uppercase string representing the plaintiff's name, the Maryland plaintiff column contained an unparsed JSON. This plaintiff JSON contained not only the plaintiff's name, but their address, business name, zip code, and more. In order to examine the Maryland plaintiffs in the same way that the Oklahoma data was examined, the JSON in that column first had to be parsed. As such, the data from the plaintiff column of the Maryland dataset was parsed into its own dataset, exemplified in table III, where each column corresponds to a key in the original JSON. As such, the "Name" column in the cleaned Maryland plaintiff dataset most closely mirrors the "plaintiff" column in the Oklahoma dataset.

Table I: Oklahoma Dataset

court	casenum	...	plaintiff	defname	issue	disp_date	disp_case	summary_date	iss_desc
STEPHENS	SC-2020-12345	...	WESTERN SHAMROCK FINANCE	DOE, JANE	OTHER	2020-09-17	DEFAULT JUDGMENT	2021-04-07	LOAN DEFAULT

762511 rows × 14 columns

Table II: Original (Unclean) Maryland Dataset

parser	versions	error	plaintiff	defendant	...	caption	caseNumber	id
districtCivil	["case-1234567-2022-1-24.html"]	True	{"City": "RICHMOND", "Name": "CAPITAL ONE BANK...	{"City": "ELKRIDGE", "Name": "JOHN, SMITH...	...	CAPITAL ONE BANK vs. SMITH, JOHN	123456789 Claim Type: CONTRACT	12345678

32534 rows × 28 columns

Table III: Maryland Plaintiff Data after parsing Plaintiff Column

City	Name	State	Address	Zip Code	Raw Data	...	Business or Organization Name	Amount	Removal Date
RICHMOND	CAPITAL ONE BANK	VA	1234 ROAD ROAD	123 456	{"City": "RICHMOND", "Name": "CAPITAL ONE BANK...	...	University Of Maryland Medical System Corporation	1234 AVENUE AVENUE	10/28/2014

34268 rows × 12 columns

Note that all sensitive data has been cleaned to protect the identities of the individuals whose cases are shown above.

Moreover, in some cases, the plaintiff column in the Maryland dataset was a list of multiple JSONs, meaning that multiple plaintiffs were related to the case at hand. As this study was simply a preliminary examination of the data, the plaintiffs linked in a JSON list were split into their own rows in the set of clean data. This choice was made so that the cleaned Maryland data would mirror the pre-cleaned Oklahoma data, which also contains cases with multiple plaintiffs, whereupon each plaintiff has their own row in the dataset. Therefore, for both the Oklahoma and Maryland datasets, the results are likely inflated if multiple plaintiffs are related to a single medical-related court case.

III. Methodology

i. Basic Medical Filtering

For the first classification method involved checking if the string corresponding to the plaintiff data contained any of the four terms found in the MEDICAL collection in table IV. These terms— “clinic”, “health”, “hospital”, and “medical”— were selected by the researchers’ common knowledge as words that would denote the plaintiff as explicitly related to the medical field. The plaintiffs were then classified through word lookup, where the plaintiff string was checked against each term in the MEDICAL collection; however, the plaintiff string on which the lookup was performed was different for each dataset. For the Oklahoma dataset, the term lookup was performed on the uppercase name of the plaintiff, as seen in the “plaintiff” column in table I. For the Maryland data, however, the word lookup was performed on the unparsed JSON string in the “plaintiff” column of table II, which corresponds to the “Raw Data” column in table III.

The reason for this discrepancy is that there is more data available in the JSON than simply in the plaintiff’s name. On some occasions a medical facility has a name that does not contain one of the four terms found in the MEDICAL collection, but that name is present in the address associated with the plaintiff. For example, a plaintiff with the name “Red River Services” located at the “456 Medical Suites” should be flagged as medical-related, as it is a service located in a medical suite; however, it would only be flagged as such in the Maryland dataset due to the richness of its data, a level of richness not reflected in the Oklahoma database.

Table IV: Collections of Medical Terminology

Collection Name	Usage	Terms
MEDICAL	Basic Medical Filtering	clinic, health, hospital, medical
omit_words	Word Embeddings	camp, care, education, environmental, hygiene, nearby, nutrition, poor, public, study, veterinary
MEDICAL_EMBED_GLOVE	Word Embeddings	aid, ambulance, center, doctor, healthcare, heart, hospice, hospitalized, medicine, nurse, nursing, outpatient, pediatric, physician, prevention, psychiatric, specialist, welfare
MEDICAL_EMBED_W2V	Word Embeddings	ambulance, dental, dispensary, doctor, gynecologist, hospitalized, immunization, infirmary, medicine, orthopedic, outpatient, patient, pharmacy, physician, reproductive, surgical, welfare, wellbeing
MEDICAL_EMBED_BOTH	Word Embeddings	ambulance, doctor, hospitalized, medicine, outpatient, physician, welfare

If the plaintiff was not determined to be related to the medical field, Named Entity Recognition was used to further classify the data. Named Entity Recognition, or NER, is the process of identifying the named entities in text and determining what kind of named entity each chunk of text is [3]. In order to perform Named Entity Recognition, one must first break the given text into chunks, or nouns and grouped noun phrases [3]. Chunking also has a prerequisite in that first the text to be chunked must be first tagged as different parts of speech. A part of speech, or POS, is a grammatical term that deals with the role words play when used in together. Tagging parts of speech, or POS tagging, is therefore the task of classifying words according to their part of speech [3]. Therefore, in order to determine a way to classify the plaintiffs that did not fall under the category of "medical-related", the text data in the plaintiff column first had to be broken into parts of speech, then chunked, and then classified via Named Entity Recognition. A list of common NER classifications can be found in table V.

Named Entity Recognition, however, is not a silver bullet; many plaintiffs in both the Oklahoma and Maryland datasets were not identified using NER, and therefore were labelled as "NO_LABEL". NER worked best on rows with relatively short strings in the plaintiff column, such as those most frequent in the Oklahoma dataset, as the data might only have one identifiable named entity. For larger strings, many entities can be identified within. In this case, the plaintiff is labelled with the mode entity, or the most frequent entity to appear in the plaintiff text data.

The result of the basic medical filtering is displayed in Table VI and will be discussed in the paper's subsequent "Results" section.

Table V: Entity Types found in Named Entity Recognition [1]

NE type	Examples
ORGANIZATION	Georgia-Pacific Corp., WHO
PERSON	Eddy Bonte, President Obama
LOCATION	Murray River, Mount Everest

DATE	June, 2008-06-29
TIME	two fifty a m, 1:30 p.m.
MONEY	175 million Canadian dollars, GBP 10.40
PERCENT	twenty pct, 18.75 %
FACILITY	Washington Monument, Stonehenge
GPE or GSP	Global Political Entity or Geographical-Social-Political Entity i.e. South East Asia

ii. Medical Filtering via Word Embeddings

The second method for classifying plaintiffs as medical-related used word embeddings to build a vocabulary of medical-related terminology to check against the data in the plaintiff column. Word embeddings are real-number vectors that represent words from a vocabulary [2]. These vectors are based on the usage of words, meaning that words that are used in similar ways have similar vectors. This both naturally captures a word's meaning and makes it easy to determine which other words are semantically similar have similar vectors, which means they map to similar locations in N-D space [2].

For the purposes of this study, word embedding was performed with prebuilt models built off both Google's Word2Vec algorithm and Stanford's Global Vectors for Word Representations (GloVe) algorithm. The Word2Vec model was built using Google News Articles, whereas the GloVe model was built using Wikipedia articles [2]. In this paper, the prebuilt Word2Vec model was imported from "pruned.word2vec" and the prebuild GloVe model was imported from "glove-wiki-gigaword-50".

Once each word embedding model was imported, medical term collections MEDICAL_EMBED_GLOVE and MEDICAL_EMBED_W2V, found in table IV, were created. To create these word sets, each collection's respective word embedding model retrieved the 10 most similar words to each of the four terms in the MEDICAL collection. The resulting similar words were first deduplicated, then lemmatized. Lemmatizing reduces words to their core meaning, and this was performed to simplify the terms and to further deduplicate them. Finally, each collection was cleansed of terms which were determined to produce false positives, as seen in the "omit_words" collection in Table IV. Once those two collections were created, the words that appeared in both collections were stored in their own set, MEDICAL_EMBED_BOTH.

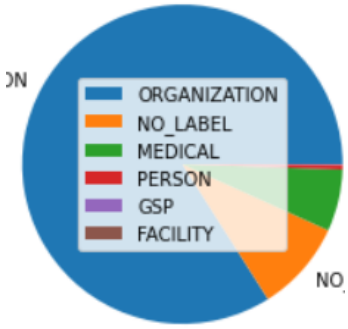
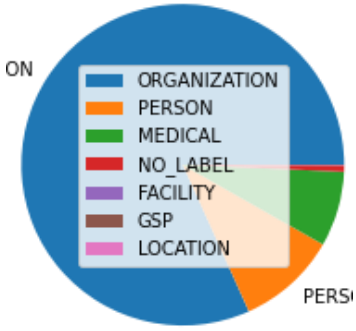
Once the three collections of terms produced by word embedding were created, the data in each dataset's plaintiff column could be checked against them, though basic filtering still applied as it was determined to be more reliable than the word embeddings. Thus, the plaintiff was first checked against basic filtering, then it was checked against MEDICAL_EMBED_BOTH, MEDICAL_EMBED_W2V, and MEDICAL_EMBED_GLOVE in that order. If the plaintiff still was not flagged as related to the medical field, the plaintiff was labelled according to Named Entity Recognition as described in the previous subsection.

The result of the medical filtering via word embeddings is displayed in Table VII and will be discussed in the paper's subsequent "Results" section.

IV. Results

i. The MEDICAL Word Set's Classification Results

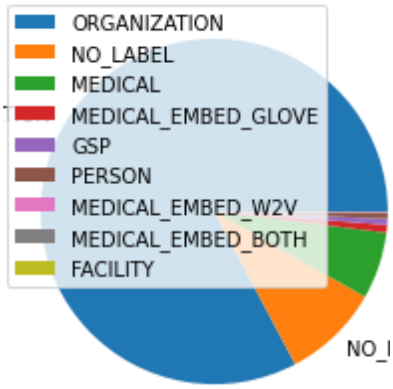
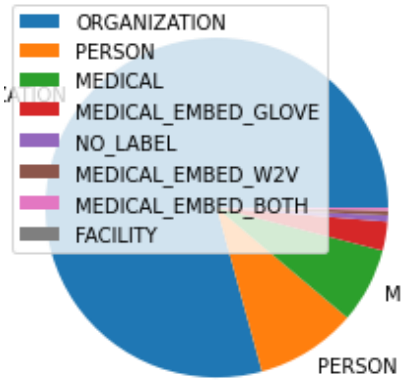
Table VI: Results of Basic Medical Filtering by Dataset

Dataset	*Counts Graph*	*Counts*
Oklahoma		ORGANIZATION 619003 NO_LABEL 68089 MEDICAL 46110 PERSON 3588 GSP 354 FACILITY 9
Maryland		ORGANIZATION 26001 PERSON 3191 MEDICAL 2403 NO_LABEL 207 FACILITY 13 GSP 1 LOCATION 1

Therefore, the figures in the table show that basic medical filtering reveals that 6.25% of the plaintiffs in the Oklahoma dataset and 7.55% of the Maryland's plaintiffs contain the words "clinic", "health", "hospital", and "medical", and were thus classified as relating to the medical field. Fewer rows of the Maryland dataset were processed than Oklahoma's— 32,534 of Maryland's as opposed to 762,511 of Oklahoma's— but more of Maryland's plaintiffs were flagged as medical-related. One can assume that the richness of information in the Maryland plaintiff column results in a greater chance of the plaintiff being classified as medical, as described in the beginning of section III.iii.

ii. Word Embeddings' Classification Results

Table VII: Results of Medical Filtering via Word Embeddings by Dataset

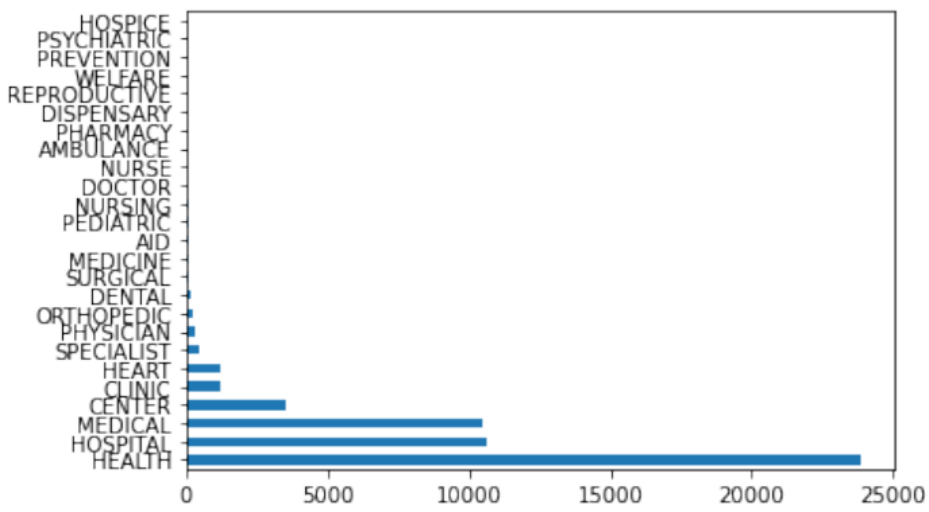
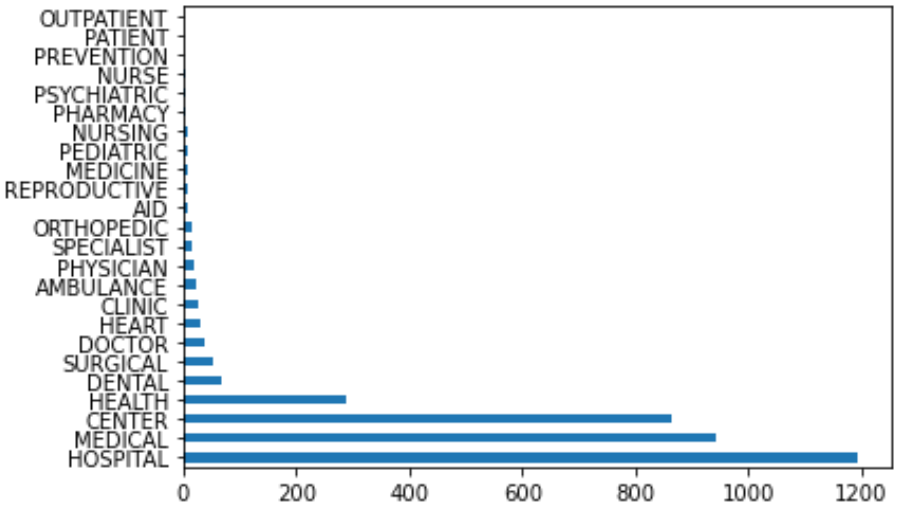
Dataset	*Counts Graph*	*Counts*
Oklahoma		ORGANIZATION 609854 NO_LABEL 66736 MEDICAL 46110 MEDICAL_EMBED_GLOVE 5344 GSP 4145 PERSON 3996 MEDICAL_EMBED_W2V 496 MEDICAL_EMBED_BOTH 446 FACILITY 26
Maryland		ORGANIZATION 27156 PERSON 3269 MEDICAL 2448 MEDICAL_EMBED_GLOVE 943 NO_LABEL 208 MEDICAL_EMBED_W2V 151 MEDICAL_EMBED_BOTH 90 FACILITY 3

The figures in the table show that filtering the plaintiff column with medical embeddings resulted in flagging an additional 0.85% of Oklahoma's and 3.72% of Maryland's plaintiffs as medical-related. As with basic filtering, the richness of Maryland's data contributes to its greater percentage of plaintiffs flagged. Therefore, when the results of the basic medical filtering are combined with those of the medical embedding, we can determine that 11.42% of Maryland's plaintiffs and 7.11% of Oklahoma's were flagged as relating to the medical field.

Moreover, both the Maryland and Oklahoma data shows that the set of words built by the GLoVe model is much more likely to flag plaintiffs as medical-related, flagging on its own 80% of the Maryland's plaintiffs discovered by word embeddings and 85% of Oklahoma's. One would expect that Word2Vec's collection of medical terms to be more effective at matching plaintiffs, as it was checked against the plaintiff before GloVe's collection; however, that was not the case with this data. That being said, one reason for this drastic superiority of the GloVe model could be due to the size of the pretrained model downloaded, as the study used a pruned version of the Word2Vec model, meaning the Word2Vec model contained fewer words and less detailed vectors. The subsequent section will elaborate on this discrepancy between the two word embedding models.

iii. Examining the Flagged Words in the Plaintiff Data

Table VIII: Most Flagged Medical Words by Dataset

Dataset	*Most Prevalent Words Flagging Plaintiff as Medical*
Oklahoma	
Maryland	

The above tables show the frequency of terms in the plaintiff column that result in the plaintiff being flagged as medical-related. Unsurprisingly, the four terms in the MEDICAL collection— “clinic”, “health”, “hospital”, and “medical”— are the in the top five flagged words for both datasets. This makes sense, as these words were checked against plaintiffs first, meaning they were more likely to flag plaintiffs in the first place.

One word that is both frequently flagged and not the MEDICAL word set is “center”; in of the Maryland plaintiffs, the word “center” is the third most flagged word, and it is the fourth most flagged word of Oklahoma’s. The term “center” was determined to be semantically similar to one of the four MEDICAL terms by the GLoVe algorithm, and it is prone to produce false positives. For example, in both datasets, daycare centers were being erroneously flagged as medical-related. Additionally, the term “center” is more frequently flagged in the dataset where more information than the plaintiff's name is being used for

medical classification. One must conclude that the greater frequency of the flag word "center" in the Maryland dataset is due to false positives created, for example, by plaintiffs located at addresses with the name "center" in it. Moreover, the word "center" is therefore the main reason why the GLoVe-generated set of words so outperforms Word2Vec', a word that is rife with false positives.

Finally, while the rest of the flagged terms flag very few plaintiffs when compared to the top seven or so words, they are more explicitly related to the medical field and provide more confident assessments of medical-related plaintiffs. Therefore, these terms provide a more accurate view of medical debt.

V. Conclusion

In this paper, plaintiffs from the civil courts in Oklahoma and Maryland were classified as being medical-related by mean of word lookup. Plaintiffs in the Oklahoma database were classified solely based on their names, whereas an unparsed JSON rich with data was used to classify Maryland's plaintiffs. This resulted in a larger proportion of Maryland's plaintiffs being flagged as medical-related, but the data checked affords greater chance at producing false positives. Moreover, the terms used in the classification of the plaintiffs were generated in two ways, the first based on the researcher's common knowledge referred to as the MEDICAL word set. The second classification method used terms semantically similar words to the MEDICAL set, as generated by two different word embedding models.

Though this paper is most confident about the flagging accuracy of the common-knowledge word set, the accuracy of the sets created by GloVe and Word2Vec word embedding algorithms is less certain. While the word embedding algorithms generated vaguer terms which could result in false positives, such as the word "center", they also produced terms like "ambulance" that are indisputably related to the medical field. Therefore, one can conclude both classification methods are effective and reasonably trustworthy; however, since the MEDICAL word set is checked before any of those produced by word embedding, it produces significantly more results.

One of the main goals of the Civil Justice Data Commons is to keep policy makers informed with the state of the civil courts in the US. Thus, the plaintiffs flagged via the two classification methods discussed in this paper provide clarity as to the proportion of civil court cases related to the medical field, which gives such policy makers the tools to examine medical debt in the US at a greater level of granularity.

i. Further Work

In future work, it is worth linking the plaintiffs to their respective cases to reduce artificial bloat of the results in the case that multiple medical-related plaintiffs brough a single case to court. Additionally, a greater amount of the data in other columns, such as Maryland's "misc" column, could be used both to flag more plaintiffs, and to increase the accuracy of those plaintiffs flagged. Finally, the strings in the plaintiff column could be lemmatized before they were checked to reduce the complexity of those words and increase the likelihood of their plaintiffs being flagged.

Moreover, different models and methods could be used to generate semantically similar words to those in the MEDICAL word set. For example, in further study a more expansive model of the Word2Vec algorithm could be used, or one could train their own model based off, say, court transcriptions or medical literature. A model such as that would, in theory, flag more plaintiffs with greater accuracy.

References

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